MILITARY SPECIFICATION

AFOCAL ASSEMBLY, AN/VSG-2

This specification is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SOPE

1.1 Scope. This specification covers the Afocal Assembly, AN/VSG-2.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallets, Material Handling, Wood Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	- Strapping, Steel, and Seals
NNN-P-40	- Paper, Lens
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-C-795	- Cushioning Material, Flexible, Cellular, Plastic Film for Packaging Applications
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	- Strapping, Nonmetallic (and Connectors)
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-97	- Tape, Packaging/Industrial, Filament Reinforced

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5855

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-I-49384	- Imager, Optical SU-121/UA

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized and Containerized Unit Load, 40 Inch x 48 Inch Pallets, Skids, Runners, or Palletized Type Base
MIL-STD-150	- Photographic Lenses
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment

DRAWINGS

DLSM-B-805093	- Afocal Assembly, AN/VSG-2

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition. The Afocal Assembly, AN/VSG-2, herein known as the afocal assembly, contains the afocal infrared optics, the manual field-of-view change mechanism, and the range focus mechanism. The afocal assembly collects the incident IR energy and relays it to the scanner.

3.2 Construction. The afocal assembly shall be constructed in accordance with DLSM-B-805093 and as specified herein.

3.3 First article. When specified in the contract (see 6.2), the contractor shall furnish three afocal assemblies for first article inspection and approval.

3.4 Parts, materials and processes. Parts, materials and processes shall be as specified herein and as shown on the applicable drawings. Parts, materials and processes not specified shall be selected in accordance with MIL-P-11268.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

3.5 Components. The afocal assembly shall consist of optical elements, mechanical parts, and a housing.

3.6 Performance characteristics. The following performance characteristics are defined for a spectral band of 7.6 to 11.75 micrometer region.

3.6.1 Effective focal length, narrow field of view (NFOV). The effective focal length shall be 10.67 inches, ±2 percent when used with the optical imager (MIL-I-49384).

3.6.2 Transmission NFOV. The average transmission of the afocal alone shall not be less than 70 percent over the spectral range of the bandpass filter.

3.6.3 Modulation transfer function (MTF). The MTF on-axis of the NFOV and wide field of view (WFOV) in the scan direction shall not be less than the values indicated in this table, when used with an optical imager:

LP/mm	MTF (%)
1.87	93
3.75	86
7.5	74
10.0	65

3.6.4 Dimensions NFOV. The afocal assembly shall have an NFOV not less than 2.5 degrees in elevation by 5 degrees in azimuth when used with the optical imager and mechanical scanner.

3.6.5 NFOV Focus. The NFOV shall be capable of displaying objects from 50 meters to infinity in sharp focus.

3.6.6 Relative illumination. The relative illumination of the NFOV/WFOV afocal assembly shall be measured with the optical imager and shall be not less than 60% at the edge of a 20 mm diameter field.

3.6.7 Distortion. The linear distortion of the NFOV/WFOV afocal assembly shall be between 4% pincushion and 4% barrel when measured with an IR imager.

3.6.8 Field flatness. The afocal assembly NFOV/WFOV shall have an off-axis MTF no less than 50% at 10 lp/mm measured in the scan direction. This measurement shall be in the same focal plan chosen for the on-axis measurement corresponding to an image translation 6 mm perpendicular to the long dimension of the detector array.

3.6.9 Effective focal length WFOV. The effective focal length shall be 3.56 inches, ±2 percent when used with the optical imager (MIL-I-49384).

3.6.10 Transmission WFOV. The average transmission of the afocal alone shall be not less than 60 percent over the spectral range of the bandpass filter.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:04Z
Check the source to verify that this is the current version before use.

3.6.11 Dimensions WFOV. The afocal assembly shall have a WFOV not less than 7.5 degrees in elevation by 15 degrees in azimuth when used with the optical imager and mechanical scanner.

3.6.12 FOV alignment. The centers of the NFOV and WFOV shall be concentric to within ±30 mils in azimuth and within ±10 mils in elevation.

3.6.13 Line-of-sight movement. While focusing the NFOV, the image of the NFOV shall not shift more than 0.1 mil in any direction regardless of direction of rotation of the focus knob.

3.6.14 Torque.

3.6.14.1 Focus control torque. The torque necessary to rotate the range focus control shall not exceed 1.0 inch-pound.

3.6.14.2 FOV switching torque. The torque necessary to rotate the FOV switching control shall not exceed 6.0 inch-pounds.

3.7 Environmental conditions.

3.7.1 Temperature shock. The afocal assembly shall not be damaged (see 6.3) by sudden changes in temperature between $-57^{\circ}C$ to $+71^{\circ}C$.

3.7.2 High temperature. The afocal assembly shall not be damaged by storage to $+71^{\circ}C$ and operation to $+52^{\circ}C$.

3.7.3 Low temperature. The afocal assembly shall not be damaged by operation to $-32^{\circ}C$ and storage to $-57^{\circ}C$.

3.7.4 Shock. With the afocal assembly mounted to a fixture it shall withstand the following shocks.

a. Basic design: 28 g's, 6 milliseconds, 1/2 sine, 18 shocks total.
b. High intensity: 65 g's, 0.75 millisecond, 1/2 sine, 18 shocks total.
c. Bench handling: The afocal assembly shall not be damaged by bench handling during normal maintenance activities.

3.7.5 Vibration. The afocal assembly shall not be damaged by vibration to the following random and superimposed sinusoidal levels for a period of 15 minutes in each of three mutually perpendicular axes.

Random: 1×10^{-3} g^2/Hz, 5 to 500 Hz

Sine:	0.26 g's, Octave 1	42 to 84 Hz	X axis
	0.78 g's, Octave 1	42 to 84 Hz	Y axis
	0.85 g's, Octave 1	42 to 84 Hz	Z axis

Source: https://assist.dla.mil -- Downloaded: 20260-22T20:04Z
Check the source to verify that this is the current version before use.

3.7.6 Altitude. The afocal assembly shall operate at altitudes up to 10,000 feet above sea level and withstand transportation at altitudes up to 40,000 feet above sea level without damage.

3.8 Preconditioning. Each afocal assembly shall be subjected to a 26.65 hour minimum preconditioning prior to delivery.

3.9 Cleanliness. The afocal assembly shall contain no foreign matter such as dirt, dust, fingerprints, moisture, etc.

3.10 Weight. The total weight of the afocal assembly shall be 6.0 pounds maximum.

3.11 Treatment and painting. Unless otherwise specified, the afocal assembly and its components and parts shall be cleaned, treated and painted in accordance with the applicable drawing.

3.12 Nameplates and product marking. Unless otherwise specified, the afocal assembly, parts, components, subassemblies, and assemblies thereof shall be marked in accordance with DLSM-B-805093.

3.13 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.5).
c. Inspection of packaging (see 4.9).

4.3 First article inspection. Unless otherwise specified in the contract, first article inspection shall be performed by the contractor.

4.3.1 Inspection. All materials, parts, processes and assemblies shall be examined for conformance to the applicable specification or drawings. Inspections shall be made using MIL-STD-105, General Inspection Level II with a 2.5 percent AQL for mechanical and a 4 percent AQL for visual inspection, except as noted in table I.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:04Z
Check the source to verify that this is the current version before use.

TABLE I. Inspections.

Inspection requirement	Percent AQL	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	2.5	3.2, 3.4, 3.5	MIL-STD-252
Weight not as specified	2.5	3.10	MIL-STD-252
Marking not as specified	2.5	3.2	
Workmanship not as specified	2.5	3.13	MIL-STD-454
Treatment and painting not as specified	2.5 1/	3.2	
Dimensions not as specified	2.5	3.2	

1/ MIL-STD-105, Inspection Level S-3 or objective quality evidence.

4.3.2 Tests. Upon successful completion of inspection specified in 4.3.1, first article afocal assemblies shall be subjected to the tests in table II. Tests shall be conducted in the order listed. Failure of any tests shall constitute first article failure.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Preconditioning	3.8	4.8
Effective focal length, NFOV	3.6.1	4.6.1
Transmission NFOV	3.6.2	4.6.2
MTF	3.6.3	4.6.3
Dimensions NFOV	3.6.4	4.6.4
NFOV Focus	3.6.5	4.6.5
Relative illumination	3.6.6	4.6.6
Distortion	3.6.7	4.6.7
Field flatness	3.6.8	4.6.8
Effective focal length WFOV	3.6.9	4.6.9
Transmission WFOV	3.6.10	4.6.10
Dimensions WFOV	3.6.11	4.6.11
FOV alignment	3.6.12	4.6.12
Line-of-sight movement	3.6.13	4.6.13
Torque	3.6.14	4.6.14
Temperature shock	3.7.1	4.7.1
High temperature	3.7.2	4.7.2
Low temperature	3.7.3	4.7.3
Shock	3.7.4	4.7.4
Vibration	3.7.5	4.7.5
Altitude	3.7.6	4.7.6

Source: https://assist.dla.mil -- Downloaded: 20260-22T20:04Z
Check the source to verify that this is the current version before use.

4.4 Inspection procedures. Unless otherwise specified herein, the afocal assembly shall be tested at an ambient temperature of 23°C, ±5°C. Linearity of photometric measurement setups shall be within 5 percent.

4.5 Quality conformance inspection.

4.5.1 Inspection. Inspection shall be as specified in 4.3.1.

4.5.2 Tests.

4.5.2.1 Group A inspection. Group A inspection shall be conducted on

specified in 4.5.1. Group A tests listed in table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Preconditioning	3.8	4.8
MTF	3.6.3	4.6.3
FOV alignment	3.6.12	4.6.12
Line-of-sight movement	3.6.13	4.6.13

4.5.2.2 Group B inspection. Group B inspection shall be conducted on afocal assemblies selected from units which have passed the tests in 4.5.2.1. Sampling of lots (see 6.2) shall be in accordance with MIL-STD-105, AQL 2.5 percent defective, inspection level S3 combined. Group B tests listed in table IV may be performed in any order.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Effective focal length, NFOV	3.6.1	4.6.1
Transmission NFOV	3.6.2	4.6.2
NFOV Focus	3.6.5	4.6.5
Relative illumination	3.6.6	4.6.6
Distortion	3.6.7	4.6.7
Field flatness	3.6.8	4.6.8
Effective focal length WFOV	3.6.9	4.6.9
Transmission WFOV	3.6.10	4.6.10
Torque	3.6.14	4.6.14

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

4.5.2.3 Group C inspection. Group C inspections shall be conducted on afocal assemblies selected from units which have passed the tests in 4.5.2.1. The sample(s) shall be tested in accordance with the inspections listed in table V. Samples shall be selected in accordance with 4.5.2.3.1. Group C tests listed in table V shall be performed in the order listed.

4.5.2.3.1 Sampling for inspection. Three afocal assemblies shall be selected by the Government for all inspections in table V whenever the monthly quantity specified in the contract starts quality conformance testing, or every 30 days, whichever occurs first. The three units are not associated with inspection lots as prescribed in MIL-STD-105. The only selection prerequisite is that the three units conform to normal production processing. The first three Group C units shall begin testing at the time the first quality conformance lot (see 4.5.2.2) begins Group B testing.

TABLE V. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Dimensions NFOV	3.6.4	4.6.4
Dimensions WFOV	3.6.11	4.6.11
Temperature shock	3.7.1	4.7.1
High temperature	3.7.2	4.7.2
Low temperature	3.7.3	4.7.3
Shock	3.7.4	4.7.4
Vibration	3.7.5	4.7.5
Altitude	3.7.6	4.7.6

4.6 Test methods. Assemblies shall be tested in accordance with all tests listed in table II and in accordance with Government-approved test procedures. Unless otherwise specified, tests shall be conducted at ambient temperature with radiation in the 7.6- to 11.75-micrometer spectral region.

4.6.1 Effective focal length, NFOV. The effective focal length of the afocal assembly with the optical imager shall be measured using either Method 1 or Method 2 or calibrated focal length method of MIL-STD-150 with suitable equipment substitution required for the specified spectral region. Failure to meet the requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Transmission. The transmission to the afocal assembly shall be measured with the optical imager using Method 6 of MIL-STD-150 within the applicable spectral region and shall be determined after dividing out the transmission of the imager alone. Failure to meet the requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Modulation transfer function (MTF). The MTF of the afocal assembly NFOV and WFOV with an optical imager shall be measured using a test set calibrated and designed to provide frequency and modulus. The MTF shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

measured either continuously or in discrete steps between 0 and 10 linepairs per millimeter. The MTF test shall be run at infinite conjugates for both narrow and wide field of view. The measurements shall be performed with a blackbody source, slit and a pyroelectric detector. A bandpass filter of 7.6 to 11.75 micrometers shall be used to limit the spectral band. An IR imager which has greater than 74 percent MTF at 10 lp/mm shall be used to image the slit onto the scanning detector. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Dimensions NFOV. Set the NFOV on an in-line fixture with an IR imager and a detector. Align this assembly to a target projected by a collimator, folded off a large mirror, adjustable in both elevation and azimuth. Rotate the mirror to the full off-axis position (first left, then right) and move the detector until the signal is received. Repeat this for the full elevation motion (both up and down). Verify that the target is detectable over both extremes of movement and that the drop in signal level is no greater than 40 percent when compared to the on-axis signal. Failure to meet the requirements of 3.6.4 shall constitute failure of this test.

4.6.5 NFOV focus. When the thermal target is in the 50-meter position during the test of 4.6.13, verify that the MTF at the resolution frequency has not dropped more than 10 percent. Failure to meet the requirements of 3.6.5 shall constitute failure of this test.

4.6.6 Relative illumination. The NFOV and WFOV relative illumination of the afocal assembly/imager shall be measured using Method 8 of MIL-STD-150 within the applicable spectral range. Failure to meet the requirements of 3.6.6 shall constitute failure of this test.

4.6.7 Distortion. The distortion of the afocal assembly (WFOV and NFOV) shall be measured using either Method 27 or Method 28 of MIL-STD-150 as applicable. Failure to meet the requirements of 3.6.7 shall constitute failure of this test.

4.6.8 Field flatness. The field flatness shall be measured for a flat field MTF in accordance with 4.6.3 for both NFOV and WFOV. Failure to meet the requirements of 3.6.8 shall constitute failure of this test.

4.6.9 Effective focal length WFOV. The effective focal length of the afocal assembly with imager shall be measured using either Method 1 or Method 2 or the calibrated focal length method of MIL-STD-150 with suitable equipment substitution required for the specified spectral region. Failure to meet the requirements of 3.6.9 shall constitute failure of this test.

4.6.10 Transmission WFOV. The transmission to the afocal shall be measured with the IR optical imager using Method 6 of MIL-STD-150 within the applicable spectral region and shall be determined after dividing out the transmission of the imager alone. Failure to meet the requirements of 3.6.10 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

4.6.11 Dimensions WFOV. Switch the afocal to WFOV and using the procedure of 4.6.4, verify the FOV. Failure to meet the requirements of 3.6.11 shall constitute failure of this test.

4.6.12 FOV alignment. Using the setup of 4.6.4, move the mirror to the axial in-line position; set the afocal in NFOV; and move the detector until the signal peaks. Switch FOV and move the detector until the signal again peaks. Record the change in position and confirm that this is less than a corresponding object space displacement of 30 mils. Repeat this for two other field points and confirm that the difference in the readings corresponds to a tilt between fields of view of less than 10 mils in object space. Finding the amount of shift of a slit target requires both a tangential and sagittal scan. Failure to meet the requirements of 3.6.12 shall constitute failure of this test.

4.6.13 Line-of-sight movement. Using the in-line fixture accurately aligned to the optical axis of the collimator, set the afocal to NFOV and center the detector on the thermal target. Move the target to the 50 meter position, focus and verify that the image has not shifted more than 0.1 mil relative to the object space. Failure to meet the requirements of 3.6.13 shall constitute failure of this test.

4.6.14 Torque. Standard torque measuring equipment capable of measuring the torque value specified shall be used for this test. An appropriate adapter shall be used with the torque measuring equipment. Failure to meet the requirements of 3.6.14 shall constitute failure of this test.

4.7 Environmental tests. Unless otherwise specified, environmental tests shall be performed in accordance with MIL-STD-810 as modified by this specification. The performance tests required during environmental tests where operation is specified by MIL-STD-810 shall be FOV alignment (4.6.12), and modulation transfer function (4.6.3). Performance tests 4.6.1 through 4.6.14 shall be run upon completion of all environmental tests.

4.7.1 Temperature shock. The afocal assembly shall be tested in accordance with MIL-STD-810, Method 503, Procedure I except that the temperature of step 2 shall be -54°C. Failure to meet requirements of 3.7.1 shall constitute failure of this test.

4.7.2 High temperature. The afocal assembly shall be tested in accordance with MIL-STD-810, Method 501, Procedure I. The afocal assembly shall be operated during the final 30 minutes of step 3. Failure to meet requirements of 3.7.1 shall constitute failure of this test.

4.7.3 Low temperature. The afocal assembly shall be tested in accordance with MIL-STD-810, Method 502, Procedure I except that the storage temperature of step 2 shall be -57°C and the low operating temperature of step 4 shall be -32°C. Failure to meet the requirements of 3.7.3 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

4.7.4 Shock. The afocal assembly shall be tested in accordance with the following procedures for basic design, high intensity, and bench handling. Failure to meet the requirements of 3.7.4 shall constitute failure of this test. Shock the afocal assembly mounted as it is in a tank:

a. Basic design: 24 g's, 6milliseconds, 1/2 sine, 18 shocks total.
b. High intensity: 65 g's, 0.75 millisecond, 1/2 sine, 18 shocks total.
c. Bench handling: The afocal assembly shall not be damaged by bench handling during normal maintenance activities.

4.7.4.1 Basic design. Subject the afocal assembly to the test specified in MIL-STD-810, Method 516.1, Procedure I as modified herein for three half-sine wave shock pulses applied in both directions along three mutually perpendicular axes. The shock pulses are shown in 4.7.4. During shock pulse verify that intermittence does not exist.

4.7.4.2 High intensity. Subject the assembly to the test specified in MIL-STD-810, Method 516.1, Procedure IV as modified herein for three half-sine wave shock pulses applied in both directions along three mutually perpendicular axes. The shock pulses are shown in 4.7.4. During shock pulsing, verify that intermittence does not exist.

4.7.4.3 Bench handling. Subject the afocal assembly to the test specified in MIL-STD-810, Method 516.1, Procedure V.

4.7.5 Vibration. The afocal assembly shall be subjected to the vibration test specified in MIL-STD-810, Method 514.1, Procedure VIII, except as follows: The afocal assembly shall be vibrated in each of 3 mutually perpendicular axes to the random and superimposed sinusoidal levels specified in 3.7.5 for 15 minutes. The appropriate periodic component will be swept up each octave in one minute and down the octave the next minute. Failure to meet the requirements of 3.7.5 shall constitute failure of this test.

4.7.6 Altitude. Place the afocal assembly in an altitude chamber and subject it to the altitude test specified in MIL-STD-810, Method 500, Procedure I, except that:

a. The assembly, in a nonoperating condition, shall be exposed to an atmospheric pressure equivalent to 40,000 feet above sea level for a period of not less than 1 hour.
b. With the chamber pressure increased to 10,000 feet above sea level, the assembly shall be subjected to the performance tests of 4.7. After exposure, subject the assembly to the performance tests of 4.7.

Failure to meet the requirements of 3.7.6 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:04Z
Check the source to verify that this is the current version before use.

4.8 Preconditioning. Each afocal assembly shall be tested in accordance with Method 102 of MIL-STD-202. The following details shall apply:

a. Test condition d except that in step 1 the temperature shall be -54°C.
b. Measurements before and after cycling are not applicable.

4.9 Inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance with requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. The afocal assembly shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. The afocal assembly shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packed method IC-1.

5.1.1.4.2 Afocal assembly, AN/VSG-2. Each afocal assembly shall be individually unit packed method III as follows: wrap each assembly with paper, lens conforming to NNN-P-40, type II and secure with tape conforming to PPP-T-45, type III. Cushion each wrapped item with material conforming to PPP-C-795, Class 2, medium thickness. Place each cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

5.2.1 Level A. A quantity of afocal assemblies, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceed 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. A quantity of afocal assemblies, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of afocal assemblies, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The afocal assembly is intended for use in the Head Assembly of the Tank Mounted Thermal Sight.

6.2 Ordering data. Procurement documents will specify, as a minimum, the following:

a. Title, number, and date of this specification.
b. Lot size, if different than specified herein.
c. Requirement to perform first article and the quantity of first article units.
d. Failure analysis, and corrective action requirements for periodic inspection.
e. AQL, if different than specified herein.
f. Requirement for first article and periodic inspection test plans.
g. Disposition of test samples. The approved first article/ periodic sample(s) shall be returned to satisfactory operation and submitted for inspection and acceptance as a production item in accordance with section 4.
h. Level A or level B preservation and packing (see section 5).
i. When rough handling test is required.
j. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.
k. Requirement that commercial packaging shall meet the requirements of MIL-STD-1188, paragraph 2(3)(b).
l. Unless otherwise specified in the contract, the Government will provide an infrared imager for testing requirements in 4.6 and 4.7.
m. If afocal assemblies are procured as part of end item assemblies, the following inspections may be waived: 4.8 in group A, and groups B and C except for 4.6.2, 4.6.8 and 4.6.10.
n. For the testing requirement of 4.8, MIL-STD-202E, Change 4 shall be used in lieu of the latest revision.
o. MIL-STD-810B shall be used for environmental tests of 4.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

6.3 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and any degradation of afocal assembly performance to values less than specified herein.

CUSTODIAN:
Army - ER

PREPARING ACTIVITY:
Army - ER

Project 5955-A166

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-A-49382(ER)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.